

Information theory and coding

Subject that studies the mathematical foundations of information processing, i.e. how to compress an information source with bits and how to send these bits through a digital communications channel. It is highly based on concepts of probability, is given in the third term of the second year and is common to the degrees of telecommunication networks and audiovisual systems. Occasionally, students of the computer science degree or the biomedical degree have taken it as an elective. The subject is structured in 9 subjects with a teaching load corresponding to 5 ECTS credits, that is to say, a total of 125 hours of dedication by the student, of which 46 are face-to-face. The main learning methods of the subject are **problem-based learning (PBL)**, **gamification (G)**, **flipped classroom (FC)** and **peer review (PR)**.

Evaluation plan

Evaluation	Competencies (Appendix 1)	Criteria	Methods (Appendix 2)	Final grade weighting
Final exam (individual)	UPF1 T3, T4, T5 E1	Correction of the result obtained Completion of the explanations Understanding of the material Correction in the written expression	Multiple-choice quiz, comprehension level and language use, according to pre-established and shared criteria	60%
Seminar deliveries (groups of 2–3 students)	UPF4 T1, T2, T3, T4 E1	Correction of the result obtained Completion of the explanations Understanding of the material Correction in the written expression	Multiple-choice quiz, comprehension level and language use, according to pre-established and shared criteria (PBL)	15%
Deliveries of personal understanding in written or audiovisual format (individual)	UPF1, UPF2, UPF4 T3, T4, T5 E1	Correction of the result obtained Completion of the explanations Understanding of the material Correction in the written expression	Evaluation of the level of understanding and use of the language according to pre-established and shared criteria	15%
Kahoots in theory classes (individual)	T1, T2 E1	Correction of the result obtained	Real-time multiple-choice gamification tool (G)	5%
Preparation of a review seminar for the entire course (groups of 4–5 students)	UPF2, UPF4, UPF5 T1, T2, T4, T5 E1	Completion of the material Correction in written expression	Peer assessment according to pre-established and shared criteria (FC+PR)	5%

Activities plan

Activity	Load	Level	Socialization	Material	Evaluation
Review material prior to theory classes: lecture notes with examples and solved problems	5 h	preparation	Individually or in groups (eg in the library or on social media)	Subject notes	
Active participation in theory classes: discussing the Kahoot questions with the surrounding classmates, answering the teacher's questions, going to the blackboard to do some examples, etc.	24 h	instruction discovery consolidation	Individual phase Group discussion phase	Explanations on the board Kahoots	Kahoots
Study notes and practice exercises: problems and examples solved in the subject notes	20 h	consolidation	Individually or in groups (eg in the library or on social media)	Notes and exercises of the subject	
Active participation in seminar classes: solving a list of problems that are distributed at the beginning of the seminar, interaction with the teacher (questions, guidance, feedback) and other students, handing in the solved problems at the end of the session	20 h	instruction discovery consolidation	Work in groups of 2–3, supervised by a teacher	Statement of the seminars	Seminar deliveries
Elaboration of 3 individual assignments of personal understanding of various topics throughout the course	30 h	discovery consolidation	Small group discussion phase (eg in the library or on social media) Individual phase of preparation of the deliveries (reports or videos)	Statement of the reports	Summary reports
Preparation of a review seminar for the entire course. In the last seminar, a random seminar prepared by one group will be assigned to another group, solved and scored according to established criteria	12 h	discovery consolidation	Work in groups of 4–5 (eg in the library or on social media)	All course material	Review seminar preparation
Final exam preparation using corrected material, tutoring with the teaching staff and subject notes	14 h	consolidation	Individually or in groups (eg in the library or on social media)	All course material	Final exam

Annex 1. List of competencies

UPF common skills

- [UPF1] Autonomy
- [UPF2] Transdisciplinary and interdisciplinary knowledge
- [UPF3] Global citizen commitment citizen
- [UPF4] Communication
- [UPF5] Digital

Transversal competences

- [T1] Working in group in international and interdisciplinary contexts
- [T2] Applying leadership and coordination and proving initiative
- [T3] Applying with flexibility and creativity the acquired knowledge and adapting it to new contexts and situations
- [T4] Communicating in academic and professional contexts orally and writing in English
- [T5] Generating new ideas and incorporating them in the daily work

Specific competences

- [E1] Modelling in an abstract way the different elements of a system of telecommunications using concepts and methods of information theory

Annex 2. Example of evaluation criteria

Grading scale for content

- 1–2: poor in content and depth of understanding
- 3–4: important information is missing, tiptoes around the main concepts
- 5–6: adequate in content and understanding, a simple story is built
- 7–8: further details are provided, relates and connects several concepts, more elaborated flow
- 9–10: goes beyond the content explained in class, further conclusions and implications of the study

Grading scale for presentation

- 1–2: absence of sentences, many mixed formats and no structure
- 3–4: no paragraphs with spelling and grammar flaws, poor format and almost no structure
- 5–6: basic use of the language, minimal basic structure
- 7–8: correct use of grammar to present the concepts, well structured, could be improved
- 9–10: excellent use of scientific language, very good in structure and text, equations and figures format